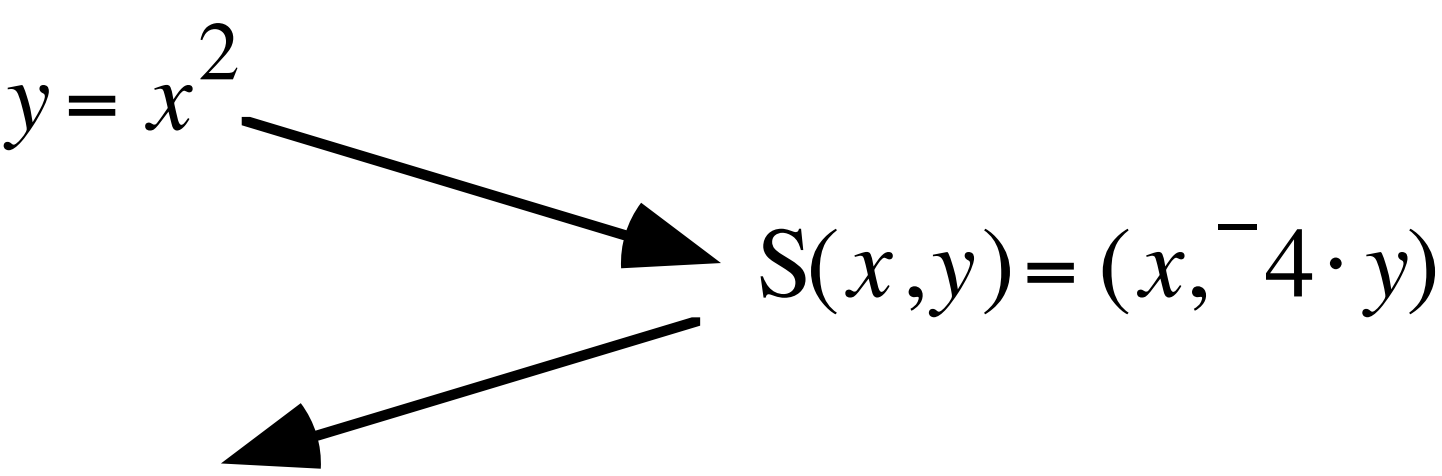
$y = x^2$
$S(x,y) = (x, -4 \cdot y)$